

Belal Ashraf Abdalhuk

Senior Software Developer

Menouf, Menoufia, Egypt | +20 127 248 4205 | belal.ash22@gmail.com
github.com/Belal2 | linkedin.com/in/belal-abdalhuk-1830b0a1

PROFESSIONAL SUMMARY

Senior Java Backend Developer with 9+ years of experience delivering enterprise platforms across telecom, legal, and geospatial surveillance domains. Specialized in building scalable microservices and real-time data pipelines with Java and Spring Boot, with a strong focus on clean architecture, security, and turning complex business requirements into reliable production systems.

WORK EXPERIENCE

Senior Backend Developer, Tahakom

02/2026 – Present

Remote Sensing Platform – geospatial surveillance platform unifying drones, satellite imagery, live video streaming, and AI-based detections.

- Designed and implemented core backend services and REST APIs for the platform using Java 21 and Spring Boot 3.
- Built real-time drone telemetry ingestion and distribution (UDP, Kafka, WebSocket) powering live map tracking.
- Integrated real-time AI detections, exposing multiple client-facing vehicle detection models over a single underlying model feeding an alerting rule engine.
- Implemented live video streaming pipelines via Oryx/SRS with configurable publish modes.
- Developed geospatial services using PostGIS and GeoTools, and managed satellite imagery and asset packages on GCS/S3.
- Secured the platform with Keycloak (OIDC) for a micro-frontend architecture; managed schema migrations with Flyway.

Technologies: Java 21, Spring Boot 3, Spring Data JPA, Hibernate, Spring Security, Keycloak (OIDC), REST APIs, Kafka, WebSocket, PostgreSQL (PostGIS), GeoTools, Temporal, Oryx/SRS, GCS, AWS S3, Flyway, Docker, Bitbucket Pipelines, GitLab, Maven.

Senior Backend Developer, STC

01/2024 – 01/2026

Inspection Management System – enterprise platform for managing inspection visits, inspectors, teams, facilities, checklists, and assignments.

- Designed and implemented backend services and REST APIs using Java 21 and Spring Boot.
- Developed the Visit Management module covering visit creation, status workflow and history, reassignment, and status-based statistics.
- Built inspection team and checklist management, linking checklists to visits and visit types and enforcing business rules such as a single team leader per team.
- Implemented dynamic search and filtering with Spring Data JPA Specifications, alongside JPQL and native PostgreSQL queries, and optimized entity loading using Entity Graphs.
- Implemented configurable lookup and validation logic using PostgreSQL JSONB.
- Secured Web and Mobile APIs with Spring Security, integrated enterprise SSO/FSSO, and configured HTTPS/SSL for Spring Boot services.
- Maintained a clean architecture (DTOs, Java Records, MapStruct mappers, validators, centralized exception handling) with OpenAPI/Swagger documentation.

Notification Management System – centralized platform for managing corporate notifications.

- Built backend services for creating, managing, and dispatching corporate notifications through email and WhatsApp channels.

Legal Case Management System – platform managing litigation filed by and against STC.

- Built APIs and modules for managing legal cases, judgments, and fines, including dashboard analytics covering case statuses, fines, and court statistics.

File Management System (Support) – enterprise document management platform.

- Supported development of services for creating, storing, converting, and retrieving files of various types, including uploads, downloads, and metadata management.
- Integrated secure storage solutions with encryption and access control, and developed RESTful and GraphQL APIs for file operations.

Technologies: Java 21, Spring Boot 3, Hibernate 6, Spring Data JPA, Spring Security, GraphQL, REST APIs, OpenAPI/Swagger, Kafka, PostgreSQL (JSONB), Resilience4j, Docker, Jenkins, Bitbucket, AWS S3, Milvus, Ollama, MapStruct, Maven, Postman, Jira.

Senior Software Developer, Lagorta**09/2022 – 12/2023**

- Assisted in the development of a dynamic advertising system, integrating with multiple messaging channels to enhance communication and advertising effectiveness.
- Collaborated with cross-functional teams to design and implement features that improved the system's performance and scalability.
- Participated in code reviews, providing constructive feedback and ensuring adherence to best practices and coding standards.

Technologies: Spring Boot, WSO2 Integration, Siddhi, PostgreSQL, MongoDB, Elasticsearch, Angular, Kubernetes, Docker, Kafka, GitLab.

Specialist Software Developer, Egabi Solutions**04/2020 – 09/2022**

- Developed integration systems connecting all external road gateways systems with the subscriptions platform.
- Developed backend modules for the Public Prosecution, Egypt Digital Archiving, and Criminal Justice systems.
- Developed the implementation and claims system for the Public Prosecutor's Office.

Technologies: Red Hat Fuse, Apache Camel, JBoss, Spring, Spring Boot, Flutter, SOA, BPM, ADF, JSF, EJB, JPA, PostgreSQL, Oracle Database, TFS, SOAP/REST web services, Git, Linux, WebLogic, Docker.

Junior & Senior Software Developer, Three Pillars**06/2017 – 03/2020**

- Developed all modules and a wide range of reports for the Arima project.
- Developed multiple projects within the Royal Saudi Air Defense program.

Technologies: Servlets, JSP, JSF, ADF, HTML, JavaScript, Jasper Reports, Oracle Database, SOAP, REST, SVN, WebLogic.

Junior Software Developer, Suez Canal University**01/2016 – 03/2017**

- Developed an IoT animal-monitoring web application.
- Developed a medical assistant application.

Technologies: Arduino, C/C++, Servlets, JSP, JSF, ADF, XML, HTML, JavaScript, Jasper Reports, Oracle Database, Firebase, SOAP, REST, SVN, WebLogic, Android, ADF Mobile.

EDUCATION

Bachelor's Degree in Computer Science, Suez Canal University**2013 – 2017****ADDITIONAL SKILLS**

- Algorithms, data structures, design patterns, systems design, and systems analysis.
- Microsoft Office (Word, Excel, PowerPoint).

References available on request.